OSSP Practical Experiments

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1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        printf("Command execution failed\n");
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed\n");
    }

    return 0;
}
```

```
aruntej@LAPTOP-DGRT8C5G:~$ nano experiment-1.c
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-1.c -o experiment-1
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-1
Enter a Linux command: ls
Parent Process
Parent PID: 757
Child PID: 763
Child Process
Child PID: 763
Parent PID: 757
Hello.c   exp2.c  experiment      experiment.c  input.txt  output.txt  program
exp1.c    exp3    experiment-1    fork          notes.txt  program1    program
exp2      exp3.c  experiment-1.c  fork.c        ossp       program1.c  program
Child process completed
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-1.c -o experiment-1
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-1
Enter a Linux command: whoami
Parent Process
Parent PID: 789
Child PID: 790
Child Process
Child PID: 790
Parent PID: 789
aruntej
Child process completed
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-1.c -o experiment-1
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-1
Enter a Linux command: pwd
Parent Process
Parent PID: 818
Child PID: 821
Child Process
Child PID: 821
Parent PID: 818
/home/aruntej
Child process completed
```

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices

```
top - 13:14:20 up 13 min,  1 user,  load average: 0.00, 0.01, 0.00
Tasks:  26 total,   1 running,  25 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0
MiB Mem :   7755.7 total,   7018.0 free,    583.1 used,    306.3 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7172.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAN
      1 root      20   0   24484  15436  11528 S   0.0   0.2   0:00.51 syster
      2 root      20   0    3180   2204   2072 S   0.0   0.0   0:00.00 init-s
      6 root      20   0    3180   2048   1940 S   0.0   0.0   0:00.00 init
     43 root      19  -1   50428  16740  15476 S   0.0   0.2   0:00.13 syster
     78 systemd+  20   0   22416  14472  11976 S   0.0   0.2   0:00.05 syster
     85 root      20   0   35424  12332   9188 S   0.0   0.2   0:00.35 syster
    104 root      20   0    2888   1952   1820 S   0.0   0.0   0:00.04 chrony
    105 root      20   0    4472   3084   2840 S   0.0   0.0   0:00.00 cron
    106 message+  20   0    8824   5460   4780 S   0.0   0.1   0:00.04 dbus-c
    118 root      20   0   44096  29804  14668 S   0.0   0.4   0:00.29 networ
    157 root      20   0   18436   9416   8176 S   0.0   0.1   0:00.09 syster
    185 syslog    20   0  220548   5868   4664 S   0.0   0.1   0:00.05 rsyslo
    225 _chrony   20   0   20424   6464   5688 S   0.0   0.1   0:00.04 chrony
    235 _chrony   20   0   12096   2440   1796 S   0.0   0.0   0:00.00 chrony
    253 root      20   0  123460  32776  18164 S   0.0   0.4   0:00.15 unatte
    287 root      20   0    5492   2660   2476 S   0.0   0.0   0:00.00 agetty
    318 root      20   0    8648   5540   4696 S   0.0   0.1   0:00.00 login
    375 aruntej   20   0   22336  13592  10980 S   0.0   0.2   0:00.07 syster
    377 aruntej   20   0   22916   4120   2116 S   0.0   0.1   0:00.00 (sd-pa
    415 aruntej   20   0    6136   5412   3864 S   0.0   0.1   0:00.02 bash
    695 root      20   0    3184   1116    980 S   0.0   0.0   0:00.00 Sessic
    696 root      20   0    3200   1256   1108 S   0.0   0.0   0:00.01 Relay(
    697 aruntej   20   0    6268   5532   3864 S   0.0   0.1   0:00.05 bash
    984 aruntej   20   0    8364   6140   3900 R   0.0   0.1   0:00.00 top
    985 root      20   0   35428   6520   3340 S   0.0   0.1   0:00.00 (udev-
    986 root      20   0   35428   6520   3340 S   0.0   0.1   0:00.00 (udev-
```